How BARCS Finds Important Genes

Answer sheet: Chapter 1 exercises

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BARCS version 0.2.1 · Reading level: grades 7 to 9

Each answer shows the calculation first. "Why this matters" explains the idea behind the question. Small differences in the last digit come from rounding.

Library	Group	K	N
1	control	150	1,000,000
2	control	210	1,400,000
3	drug	60	800,000
4	drug	45	600,000

Part A: From counts to a change

1. (a) Library 1: $150 \div 1{,}000{,}000 = 0.000150 = 150$ per million. Library 2: $210 \div 1{,}400{,}000 = 150$ per million. Library 3: $60 \div 800{,}000 = 75$ per million. Library 4: $45 \div 600{,}000 = 75$ per million.

(b) Control: $(150 + 210) \div (1{,}000{,}000 + 1{,}400{,}000) = 360 \div 2{,}400{,}000 = 150$ per million. Drug: $(60 + 45) \div 1{,}400{,}000 = 105 \div 1{,}400{,}000 = 75$ per million.

(c) Fold change: $75 \div 150 = 0.5$. $\log_2(0.5) = -1$: one halving.

Why this matters. The proportion removes the size of the library. Then libraries of different sizes can be compared.

2. The libraries have different sizes. Library 2 has 1.4 million reads and library 3 has 0.8 million. A bigger library gives bigger counts for every guide, even with no biology at all. Per million reads, library 2 has 150 and library 3 has 75. So the drop is $150 \div 75 = 2$ times, not 3.5 times. The extra $3.5 \div 2 = 1.75$ is just the difference in library size ($1.4 \div 0.8 = 1.75$).

Why this matters. Raw counts mix two things: the biology and the depth of reading. Dividing by N takes the depth out.

3. (a) $0.00015 \div 0.99985 = 0.000150023$. $\ln(0.000150023) = -8.8047$.

(b) Reason 1: a proportion must stay between 0 and 1, but a log-odds can be any number. So BARCS can add effects ($\beta_0 + \beta_1 + \dots$) without ever giving an impossible proportion. Reason 2: on a log scale, "half" and "double" are the same size with opposite signs: $\ln(0.5) = -0.693$ and $\ln(2) = +0.693$. Changes that multiply become changes that add, which is what the design matrix does.

Why this matters. The design matrix adds up β values. That only makes sense on a scale where effects add and no value is forbidden.

Part B: Noise

4. (a) Distances: -10, 0, $+10$, 0. Sum: $-10 + 0 + 10 + 0 = 0$.

(b) Squares: 100, 0, 100, 0. Average: $200 \div 4 = 50$. Standard deviation: $\sqrt{50} = 7.07$.

(c) The plain distances always add up to 0, because the values above the average balance the values below it. So they cannot measure the spread. Squares are never negative, so they do not cancel. A second reason: squared spreads of independent values add up, and BARCS needs that.

Why this matters. Every noise formula in BARCS is a variance. This is the reason the variance uses squares.

5. (a) $1{,}000{,}000 \times 0.00015 \times 0.99985 = 149.98 \approx 150$. Standard deviation: $\sqrt{150} = 12.2$ reads.

(b) Extra factor: $1 + 999{,}999 \times 0.000001 = 1.999999 \approx 2.0$. Total variance: $150 \times 2.0 = 300$. Standard deviation: $\sqrt{300} = 17.3$ reads.

(c) $1{,}000{,}000 \times 999{,}999 \times 0.000001 \times 0.00015 \times 0.99985 = 149.98 \approx 150$. Check: $150 + 150 = 300$. Correct.

Why this matters. Here the biological noise is as large as the counting noise. A model with counting noise only would make the noise look half as big as it is, and too many guides would look significant.

6. (a) Two reads from the same library have the correlation 0.000001. If one read is guide g7, the next read is very slightly more likely to be g7 too. The correlation of one pair is tiny. But a library of a million reads has almost a million million pairs, so together they double the variance (Exercise 5).

(b) $\rho = 0$. If every flask has exactly the same true proportion, there is no wobble, and the reads are unrelated.

Why this matters. ρ is not a free knob. It is a correlation that you can measure, and it comes from flasks that are truly different.

7. (a) A–B, A–C, A–D, B–A, B–C, B–D, C–A, C–B, C–D, D–A, D–B, D–C. That is 12 pairs.

(b) $4 \times 3 = 12$. Correct.

(c) $N \times N = 16$ would also count A–A, B–B, C–C and D–D. A read does not form a pair with itself. Those four "self" terms are the squares, and they are the counting noise. They are already counted in $N \times \mu \times (1 - \mu)$.

Why this matters. The variance of a count is "N squares + $N \times (N - 1)$ pairs". The squares give counting noise, the pairs give biological noise.

8. (a) $N = 4$ million: variance $= 599.9 \times (1 + 3{,}999{,}999 \times 0.000001) = 599.9 \times 5.0 = 2{,}999.5$; SD $= 54.8$ reads; $54.8 \div 4 = 13.7$ per million.
$N = 2$ million: variance $= 299.96 \times 3.0 = 899.9$; SD $= 30.0$ reads; $30.0 \div 2 = 15.0$ per million.
$N = 1$ million: variance $= 300$; SD $= 17.3$ reads; 17.3 per million.

(b) A: $13.7 \div \sqrt{2} = 9.7$. B: $15.0 \div \sqrt{4} = 7.5$. C: $17.3 \div \sqrt{8} = 6.1$ per million.

(c) Option C is the most precise. The extra factor grows with N (2, 3 and 5 for 1, 2 and 4 million reads). Deeper reading shrinks the counting noise, but not the biological noise. One library can never be more precise than $\sqrt{\rho \times \mu \times (1 - \mu)} = 12.2$ per million, however deep you read. Only more replicates average the biological noise away. More replicates also give more degrees of freedom for the test.

Why this matters. This is the most practical lesson of Chapter 1: when biological noise is present, spend money on replicates, not on depth.

Part C: The design matrix

9.

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \qquad \text{df} = 6 - 2 = 4.$$

Why this matters. Each extra replicate adds a row but no column, so it adds one degree of freedom.

10. (a)

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \quad \begin{matrix} \text{donor 1 control} \\ \text{donor 1 drug} \\ \text{donor 2 control} \\ \text{donor 2 drug} \end{matrix}$$

(b) The "donor 2" column takes away the difference between the two donors. Then the drug effect is measured inside each donor. Without it, the donor difference counts as noise. The SE becomes bigger, and the test can miss a real effect. If the donors were not balanced, the donor difference could even look like a drug effect.

(c) $\text{df} = 4 - 3 = 1$. One degree of freedom gives very little information about the noise. The test needs $|t| > 12.71$ for $p < 0.05$. More donors would help.

Why this matters. Every column must have a reason. A column for a known difference removes noise, but it also costs one degree of freedom.

11. (a) $\eta = 1 \times (-8.8048) + 1 \times (-0.6931) = -9.4979$.

(b) $e^{-9.4979} = 0.0000750$. $\mu = 0.0000750 \div 1.0000750 = 0.0000750 = 75$ per million.

(c) $800{,}000 \times 0.0000750 = 60$ reads. The real count is 60. They agree.

Why this matters. A row of X times the β values gives a prediction for any library. This is how BARCS compares the model with the data.

12. $e^{-0.6931} = 0.50$. The drug halves the odds of guide g7. For small proportions the odds and the proportion are almost equal, so the proportion also halves (150 to 75 per million). On the log-odds scale the effect is the same number for any starting level, and it adds to β_0 in the design matrix.

Why this matters. β_1 is the answer to the biological question: by how much does the drug change this guide?